← Back **Quiz 4**
Graded Assignment • 1h

Due Mar 25, 11:59 PM CET

Your grade: 90%

Your latest: **90%** • Your highest: **90%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. What is a Python package? 1 / 1 point

- A tool used for converting Python code into other programming languages.
- Python code written by a third-party that can be downloaded and installed to enhance Python programs.
- A pre-installed library in Python that handles basic arithmetic operations.
- A built-in Python feature that automatically optimizes code performance.

Correct
A Python package is a collection of third-party code that developers can download and install to extend the functionality of their Python applications.

2. **True or False**: The following command can be used to download a python package directly inside a Jupyter notebook cell. 1 / 1 point

```
!pip install package_name
```

- True
- False

Correct
The command `!pip install package_name` is used in a Jupyter notebook to install a Python package directly. The exclamation mark `!` (bang) allows you to execute shell commands within the notebook.

3. When you run the following code in your Jupyter Notebook: 1 / 1 point

```
from helper_functions import celsius_to_fahrenheit
```

You are asking Python to load the function **celsius_to_fahrenheit** from the file **helper_functions.py**. But in order for this code to successfully execute as it is, which of the following needs to be true?

(Select all that apply)

- [ ] The "helper_functions.py" file can be anywhere on the computer.
- [x] The "helper_functions.py" file needs to be in the same directory as the Jupyter notebook.

Correct
In Python, when you import a module using a statement like "from helper_functions import celsius_to_fahrenheit", Python looks for the module in its current directory (the directory where the script or notebook is located). If the file "helper_functions.py" is in the same directory as your Jupyter notebook, Python can easily find and load the function "celsius_to_fahrenheit" from it.

- [ ] The "helper_functions.py" file needs to be in a folder one level above the Jupyter notebook.
- [x] The function **celsius_to_farenheit** must be defined in the "helper_functions.py" file.

Correct
Python will look for a function called celsius_to_farenheit inside the helper_functions.py file (assuming the file is in the correct location). If Python does not find a function named celsius_to_farenheit, the import command will return an error.

4. What is the difference between the following code lines? 1 / 1 point

```
import file_name
```

and:

```
from file_name import *
```

- "import file_name" imports the file_name module as a whole, while "from file_name import * " imports all functions and variables from the file_name module.
- "import file_name" imports all functions and variables from the file_name module, while "from file_name import * " imports only the specified functions and variables.
- "import file_name" imports only the specified functions and variables from the file_name module, while "from file_name import * " imports the file_name module as a whole.
- " import file_name" imports the file_name module as a whole, while "from file_name import * " imports only the specified functions and variables.

Correct
"import file_name" imports the entire module as a single entity, so you use it with the module name prefix, like "file_name.function_name()". "from file_name import * " imports all functions and variables from the module directly, so you can use them without any prefix.

5. **True or False**: Is this line of code correct? 1 point

```
import pandas as pan
```

- True
- False. It should be "import pandas as pd"

Incorrect
This answer is incorrect because while it is common practice to use "import pandas as pd" (since "pd" is a widely recognized abbreviation for "pandas" in the Python community), it is not a requirement. The alias can be anything you choose, as long as it follows Python's naming conventions. The line "import pandas as pan" is perfectly valid and will work correctly in your code. The choice of alias is up to the programmer and can be customized based on personal or project-specific preferences.

6. You are working with a chatbot to write code to visualize some data about used cars sales. Here is the data, which is stored as a special datatype called a pandas dataframe: 1 / 1 point

```
In [2]: data = pd.read_csv('used_cars_information.csv')
        print(data)

                    Model    Price  Year  Kilometer
0  Toyota Corolla Altis 1.8G  6000.0  2015      41000
1        Toyota Camry Hybrid 12000.0  2018      30000
2            Toyota Yaris J   3500.0  2017      40000
3 Toyota Fortuner 2.8 4x4 AT 25000.0  2019      20000
4        Toyota Etios Liva V  5000.0  2016      55000
```

Which of the following prompts would be **most effective** at having the chatbot write code to visualize your dataset as a **scatterplot**?

- Plot my data.
- I have some used car sales data stored as Pandas DataFrame. Please provide Python code using matplotlib.pyplot and Pandas only to display this data using a scatter plot. x axis should show the number of kilometers driven by each car and y axis should shows the price of each car in USD. The title of the plot should be "Car Price vs. Kilometers Driven".
- I have some used car sales data. Please provide Python code using matplotlib.pyplot and Pandas only to display this data.
- Please provide Python code using matplotlib.pyplot and Pandas only to display data about used cars using a scatter plot. The title of the plot should be "Car Price vs. Kilometers Driven".

Correct
This option is the best prompt as it provides the most detail: it asks for a "scatter plot", specifies how the data is stored, and gives detailed instructions of the packages to use and the format of the final visualization.

7. What does **API** stand for? 1 / 1 point

- Application Program Installer
- Advanced Protocol Interaction
- Automated Process Integration
- Application Programming Interface

Correct
Yes, API stands for Application Programming Interface.

8. What is the primary purpose of using APIs in software development? 1 / 1 point

- To reduce the memory usage of applications
- To enable different software applications to communicate and interact with each other.
- To enhance the security of software applications.
- To improve the speed of software development.

Correct
The primary purpose of using APIs (Application Programming Interfaces) is to allow different software applications to communicate and interact with each other. This enables developers to access functionality or data from other services without needing to understand their internal workings.

9. LLMs have a feature called **temperature**. How does adjusting the **temperature** affect the response of the language model? 1 / 1 point

- It adjusts the processing power required by the model.
- It changes the language model's ability to understand grammar.
- It controls the randomness of the response.
- It sets the model's preference for longer responses.

Correct
The temperature setting in language models determines the level of randomness in the responses, with higher temperatures producing more varied and creative outputs.

10. LLMs use a "system message" feature to guide their responses. What role does the "system message" **content** play in shaping the LLM's response? 1 / 1 point

- It determines the speed at which the LLM generates responses.
- It controls the level of detail in the LLM's answers.
- It sets the maximum length of the LLM's responses.
- It instructs the LLM on how to behave or respond.

Correct
The "system message" **content** directs the LLM on the desired manner or style of response, effectively shaping its behavior during interaction.